Biomarker Identification involved in Pancreatic Ductal Adenocarcinoma using NGS pipeline and Machine Learning Algorithms

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DECLARATION

We hereby declare that the contents of the report "**Biomarker Identification involved in Pancreatic Ductal Adenocarcinoma using NGS pipeline and Machine Learning Algorithms**" are project of our own research and no part has been copied from any published source (except the references). We further declare that this work has not been submitted for award of any other diploma/degree. The university may take action if the information provided is found false at any stage. In case of any default the scholar will be proceeded against as per UAF policy.

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) remains one of the most fatal cancers globally, with poor survival rates due to late-stage diagnosis and restricted treatment choices. Pancreatic cancer is the fourth most common cause of cancer-related deaths. In this study, we used a multifaceted strategy that included Next Generation Sequencing (NGS), RNA-seq analysis, and machine learning techniques to better understand the molecular landscape of PDAC and discover possible biomarkers and therapeutic targets. Comprehensive searches of publically available datasets provided human-origin samples from PDAC patients and nearby healthy pancreatic tissue, which were rigorously analyzed and preprocessed for data quality. Differential expression analysis found genes with significant expression changes in PDAC samples compared to normal controls, with machine learning techniques assisting with feature selection and classification. Our methodology allowed for the identification of key genetic alterations and signaling pathways associated with PDAC pathogenesis, offering promising avenues for the development of targeted therapies and precision medicine approaches. Our research enabled the identification of critical genetic changes and signaling pathways linked to PDAC pathogenesis, opening up new opportunities for the development of targeted therapeutics and precision medicine methods. Through collaborative efforts and interdisciplinary research, we aim to translate these findings into tangible clinical advancements that improve patient outcomes and quality of life. This study represents a crucial step forward in the quest to unravel the complexities of PDAC and develop effective strategies for its prevention, diagnosis, and treatment, with the ultimate goal of transforming the landscape of pancreatic cancer care and saving lives. Biomarker identification was improved by machine learning techniques, which found important genes like PRF1, ZNF200, and CD2, which are essential for differentiating PDAC from normal samples. These results shed light on the pathophysiology of PDAC and suggest possible diagnostic markers. The hazards of overfitting, the requirement for large training datasets, and the interpretability of intricate models are some of the study's shortcomings. Furthermore restricting causal inference and generalizability is the study's retrospective design.

Contents

Abstract 5
Chapter 1: Introduction 9
Chapter 2: Review of Literature 13
Chapter 3: Methodology 18
Inclusion criteria 18
Bulk RNA-Seq Analysis 18
3.1 Dataset Collection 18
3.2 Data Quality Evaluation & Preprocessing 18
3.3 Identification of differentially expressed genes (DEGs) 19
3.4 Preprocessing and Genome Alignment 19
3.5 Raw Data Quality 20
3.6 Quality Control 20
3.7 Second quality check of pre-processed data 20
3.8 Reference genome mapping 21
3.9 Read Duplication Identification and Removal 21
3.10 Assembly and Quantification of Transcripts 21
3.11 Hub Gene Identification using Machine Learning 21
3.11.1 Data Loading 21
3.11.2 Creating Level Factor 22
3.11.3 Filtering Low Counts 22
3.11.4 Normalization 22
3.11.5 Defining Custom Score and Filter Functions 22
3.11.6 Decision Tree Model (C 5.0) 22
3.11.7 Random Forest Model 22
3.11.8 Results Analysis 22
Chapter 4: Results 23
4.1 Next Generation Sequencing Analysis 23
4.2 Pre-processing of the Transcript Data 23
4.3 Visualization of DEGs 27
4.4 Machine Learning 28
4.4.1 Data Preparation 28
4.4.2 Feature Selection 28

4.4.3 Model Training and Evaluation: Decision Tree Approach 29
4.4.4 Performance Comparison 30
4.5 Identified Hub Genes 32
Chapter 5: Discussion 33
Chapter 6: Summary 35
References 36

List of Figures

Figure 1: Distribution of transcripts per gene …… Page #23

Figure 2: Regularize Log Transformation …..…… Page #24

Figure 3: Box plot of regularize log transformation …..…… Page #25

Figure 4: Dispersion Plot of normalized counts …… Page #26

Figure 5: Scatter plot of regularize log transformation ……..…… Page #27

Figure 6: Volcano plot of Differentially Expressed Genes ……..……...…… Page #27

Figure 7: Predicted significant features through machine learning model Decision Tree..Page #28

Figure 8: Confusion matrix and Statistics of Decision Tree Model ……..…..Page #29

Figure 9: Machine Learning Models Performance Comparison on PDAC Samples …….Page #*30*

Figure 10: Box Plot Performance Comparison of Machine Learning Models for PDAC Classification ……..…… Page #31

Chapter 1: Introduction

The fourth most common cause of cancer-related deaths is pancreatic cancer. According to Kirby et al. (2016), this disease develops when a pancreatic cell sustains damage that sends the malignant (cancer) cell spiraling out of control. According to GLOBOCAN 2018 predictions, there will be a significant increase in pancreatic cancer mortality (79.9%; 3,45,181 deaths) and incidence (77.7%; 3,56,358 new cases) between 2018 to 2040. This is because over the past 50 years, preclinical and clinical research has had a negligible impact on patient outcomes and has not been able to improve preventative and treatment strategies. By developing new treatments and implementing primary and secondary preventative research, it is necessary to lower the incidence and death of pancreatic cancer (Casolino et al., 2021).

There are two subtypes of pancreatic cancer: exocrine and neuroendocrine. In over 85% of cases, exocrine pancreatic ductal adenocarcinoma is the most common kind of pancreatic cancer (Lu et al., 2017). Age 65 and beyond is associated with an increased risk of pancreatic cancer (Kirby et al., 2016). A number of inherited and acquired genetic events lead to pancreatic cancer. Both high and low penetrance inherited genetic changes have a significant impact on it. Individual treatment may be possible for patients with pancreatic hereditary mutations since they may respond better to tailored medications (Chen et al., 2017). Patients with locally advanced cancer have a typical survival time of 8 to 12 months, according to Lu et al. (2017); on the other hand, patients with distant metastases have a worse prognosis, with a three to six months of median survival time.

The build-up of these genetic alterations causes significant disruptions to signaling networks, which leads to the emergence of a malignant phenotype. Patients with pancreatic ductal cancer often have between 26 and 101 non-synonymous somatic mutations, which impact critical signaling pathways and increase the malignant activity of pancreatic ductal adenocarcinoma cells. Somatic mutations in pancreatic ductal adenocarcinoma affect cell growth regulators, DNA repair systems, and cell interaction with the surrounding environment. Structural modifications have been identified by single nucleotide variants that are distributed, rearranged, unstable, and stable. Four distinct mutational alterations observed in more than 100 pancreatic ductal adenocarcinomas are related to age, APOBEC cytidine deaminases, mismatch repair deficiencies, and BRCA2 (Felsenstein et al., 2018).

The most common kind of pancreatic cancer, pancreatic ductal adenocarcinoma (PDAC), has a 5-year survival rate of less than 5%, making it one of the deadliest malignancies worldwide due to delayed discovery (Fesinmeyer et al., 2005). Differentiating pancreatic ductal adenocarcinoma from other non-malignant benign gastrointestinal disorders, like chronic pancreatitis, poses a problem in the early diagnosis of PDAC (Brand and Matamoros, 1998). Even though the sensitivity of PDAC identification has increased due to imaging tools like FDG-PET and endoscopic ultrasound, in more than 50% of cases PC cannot be distinguished from focal mass-forming pancreatitis. Early locoregional recurrence, rapid metastatic growth, inadequate treatment regimens, early asymptomatic stages, and a deficiency of clinically significant biomarkers that can detect PC in its precursor forms are all contributing factors to PC's poor

prognosis (Ballehaninna and Chamberlain, 2012). According to research, unintentionally detected stage I pancreatic tumors that are still localized to the pancreas may have a seventy percent, fiver year survival rate (Frena, 2001; Schneider and Schulze, 2003).

The development of treatment approaches that could treat the illness, increase the likelihood of survival without disease, and enhance the quality of life for patients depends on the early detection of malignancies. When necessary, definitive, frequently curative therapies including ablation techniques, surgery, or three-dimensional intensity-modulated radiation will come after systemic chemotherapy. With high-throughput RNA sequencing, or RNA-seq, all genes and their isoforms may be fully annotated and quantified across datasets, providing a comprehensive picture of the transcriptome. Compared to microarrays, RNA-seq offers lower technical variability, a larger detection range, higher resolution, and the potential to analyze new transcripts (Corchete et al., 2020).

Thus, it only makes sense to start actively testing for PDAC as soon as possible. Some of its downsides include poor specificity, lack of expression in the Lewis negative phenotype, and a higher rate of false positive elevation in the presence of obstructive jaundice. The only blood-based biomarker with FDA approval for PC diagnosis, prognosis, and management is CA19-9, which is also the most widely utilized (Ballehaninna and Chamberlain, 2012). Numerous possible biomarkers for PC management have been identified, including cytokeratin, glycoprotein, Mucinic indicators, PC-associated protein markers, and hepatocarcinoma-intestine-pancreas protein (Ballehaninna and Chamberlain, 2013). However, in the validation cohorts, none of these have been able to exceed CA19-9. Previously, researchers used a novel meta-analysis strategy on several transcriptome studies to identify a unique tissue biomarker based on five genes for the diagnosis of PDAC (Ballehaninna and Chamberlain, 2013).

At the tissue level, this biomarker panel was able to discriminate PDAC from non-tumor from PDAC precursors, as well as from CP, other malignancies, and healthy controls with eighty nine percent specificity and ninety four percent sensitivity (Bhasin et al., 2016). The challenges associated with getting biopsy samples make it unclear how relevant tissue-based diagnostic indicators are. Furthermore, the majority of recent research rely on small sample sizes, which limits their ability to find reliable biomarkers. The main unmet need for early PDAC diagnosis is the availability of trustworthy blood-based biomarkers, given the unpredictable nature of PC.

Numerous studies at genomic level, describing the molecular structure of PDAC have been motivated by the critical need for improved PDAC diagnostics in order to discover new treatment targets and biomarkers for early diagnosis. We still lack a trustworthy biomarker that can outperform CA19-9 in terms of sensitivity and specificity, despite numerous genomics research. Due to datasets originating from various batches, platforms, procedures, and analytical methods, as well as a lack of concordance, the independent research are hampered by inherent statistical limitations (Ramasamy et al., 2008). Due to differences in analytical methodology, the published gene signatures of individual microarray investigations do not accord with meta-analysis and comparative analysis research when standard procedures are carried out (Ramasamy et al., 2008).

Improved PDAC biomarkers are desperately needed, as PDAC claims the lives of 330,000 people annually worldwide. Many of these people also have to deal with the unknowns related to diagnostic tests, PDAC risk factors, and the possibility of cancer from unintentionally detected pancreatic lesions. For instance, it is frequently difficult to make a firm diagnosis of PDAC due to the limitations of cytologic evaluation of pancreatic mass lesions, particularly in cases when an on-site cytopathologist is not available and chronic pancreatitis is present (Varadarajulu et al., 2005; Nguyen et al., 2009). Due to the rapid improvements in imaging quality and volume, there has been an increase in the number of imaging procedures performed annually in the US. This has resulted in a rise in the identification of potential precursor lesions for PDAC, such as mucinous cystic neoplasms and intraductal papillary mucinous neoplasms.

A high percentage of morbidity and mortality might result after surgery, and it's frequently uncertain which lesions will progress to aggressive malignancy, even if removing precursor lesions is linked to a higher chance of survival (Grützmann et al., 2010). For these individuals, an accurate biomarker to enable risk classification would significantly improve the current diagnostic and decision-making dilemma. Reliable biomarkers are crucial for enhancing screening, according to Ryan, Hong, and Bardeesy (2014). This is especially true for individuals who may be more vulnerable to PDAC, such as those with a family history of the illness, type 3 diabetes, genetic disorders, BRCA2 carriers, chronic pancreatitis, and smokers.

All of the PDAC versus healthy patient gene expression datasets from the GEO and ArrayExpress databases2, which were evaluated using microarray or sequencing platforms, are included in our analysis. The datasets obtained from blood and tissue sources that were excluded are now included. The methodology employed in this study encompassed a multifaceted approach integrating Next Generation Sequencing (NGS), RNA-seq analysis, and machine learning techniques. The initial data gathering involves extensive searches of publicly available datasets from sources such as Array Express and EMBL-EBI, with an emphasis on pancreatic cancer and pancreatic ductal adenocarcinoma (PDAC). The selection criteria were severe, with a focus on human-origin samples from patients with PDAC and nearby healthy pancreatic tissue, while rejecting cell line-based research and samples with therapy-induced changes. Bulk RNA-seq analysis was performed on the obtained datasets, with strict data quality evaluation and preprocessing steps to reduce technical noise and enable robust downstream analysis. Advanced statistical techniques such as linear regression and Bayesian inference were used to identify genes with significant expression differences in PDAC samples compared to normal controls. Furthermore, machine learning methods such as decision trees and random forests were used to identify features and classify data based on gene expression profiles. We used this comprehensive methodology to explore the molecular pathways behind PDAC pathogenesis and discover prospective biomarkers and therapeutic targets with clinical significance.

In conclusion, this study emphasizes the critical need for novel ways to battle pancreatic ductal adenocarcinoma (PDAC), one of the most lethal cancers with dangerously poor survival rates. By dissecting the molecular landscape of PDAC using Next Generation Sequencing (NGS) and

RNA-seq analysis, we hope to discover novel biomarkers and therapeutic targets that will transform early diagnosis and treatment techniques. The identification of important genetic abnormalities and signaling pathways linked to PDAC pathogenesis opens up new opportunities for the development of tailored treatments and precision medicine techniques.

Furthermore, our use of machine learning techniques improves the predictive ability of our findings, making it easier to identify key gene signatures and hub genes with diagnostic and prognostic implications. We hope to transform these results into practical clinical advances that improve patient outcomes and quality of life through joint efforts and multidisciplinary research. By addressing the limits of current diagnostic and treatment approaches, our study hopes to contribute to the ongoing battle against PDAC, providing hope for a brighter future in the fight against this dreadful disease. It is critical that we continue to investigate novel methodologies and develop collaboration across scientific disciplines in order to accelerate the translation of research findings into clinical practice. Furthermore, the findings from our study have the potential to not only improve PDAC management, but also to inform preventive efforts and public health programs targeted at minimizing the disease's impact. By leveraging genetics, bioinformatics, and machine learning, we can pave the road for individualized cancer care that prioritizes early detection, focused interventions, and better patient outcomes. In summary, our work is an important step forward in understanding the intricacies of PDAC and developing better preventive, diagnosis, and treatment techniques. With continuous investment in research and innovation, we remain hopeful about the potential to revolutionize the landscape of pancreatic cancer care and, ultimately, save lives.

Chapter 2: Review of Literature

For pancreatic adenocarcinoma, a diagnosis made before the age of 40 is rare, with an average age of 71 years old. Worldwide, 1–10 people per 100,000 are affected with pancreatic cancer, of which 85% have adenocarcinomas. According to Jemal et al. (2011), pancreatic cancer has remained higher in men and developed nations over the past 30 years than other prevalent solid tumors. In the world, it ranks as the eighth most common cause of cancer-related death for men and the ninth most common cause for women. This year, 46,000 Americans are predicted to develop pancreatic cancer, and 40,000 are predicted to lose their lives to the disease (Siegel et al., 2014). While 5–10% of pancreatic tumors are thought to have an inherited component, most instances lack a known genetic cause for family aggregation (Klein et al., 2004). The relative risk of acquiring pancreatic cancer increases in a dose-dependent manner in those with a verified family history of the disease in a first-degree relative. Those with one, two, or three first-degree relatives who are affected have a relative risk increase of two, six, or thirty times higher than the general population (Jacobs et al., 2010).

The high rate of activating mutations in KRAS (more than ninety percent), the progression from different precursor lesions, the propensity for both local invasion and distant metastasis, the extensive stromal reaction that generates a hypoxic and hypo-vascular microenvironment, the reprogramming of cellular metabolism, and the evasion of tumor immunity are some of the characteristics that establish pancreatic adenocarcinoma (Feig et al., 2012). A model of the development of pancreatic adenocarcinoma has been constructed by investigations into molecular pathology and comprehensive genomic analyses. Microscopic premalignant pancreatic lesions connected to the pancreatic ducts have been found to harbor the characteristic mutations of pancreatic adenocarcinoma; these observations are known as pancreatic intraepithelial neoplasia. About 10 to 15% of pancreatic adenocarcinomas show a stepwise progression, similar to the model developed for the polyp-to-adenocarcinoma sequence in colon cancer, along with other, less common changes (Biankin et al., 2012). Since these are changes to tumor suppressor genes, no treatment intervention theories have been developed as of yet. The pancreatic adenocarcinoma genome is distinguished by a variety of extensive chromosomal alterations, including amplifications, deletions, and rearrangements, in addition to these mutational events. Another kind of precursor to pancreatic cancer is an intraductal papillary mucinous tumor. These radiographically visible cystic tumors have a varied histologic and clinical appearance and are very frequent, occurring in 2% of adults and up to 10% of people aged 70 or beyond. The majority of intraductal papillary mucinous neoplasms are asymptomatic, but they carry a 25% risk of invasive cancer overall. Additionally, tumors originating from the main pancreatic duct are far more likely to become malignant than those originating from the branch duct. Similar to pancreatic epithelial neoplasms, KRAS mutations are commonly found in intraductal papillary mucinous neoplasms (40–65% of cases) (Wu et al., 2011). Though it only makes up around 15% of cases, pancreatic adenocarcinoma that has been removed has been the subject of most genetic investigations. As has been seen in other cancer types, exome sequencing from a fast autopsy program, in which patients with pancreatic cancer agreed to an autopsy being conducted as soon

as they died, has revealed significant intra-tumoral genomic heterogeneity, with multiple unique subclones coming from a common originating cell (Iacobuzio-Donahue et al., 2009). The primary tumors and the metastatic deposits removed from the same patient share a substantial amount of genetic information. However, metastases originate from distinct subclones inside the initial tumor.

Furthermore, different genetic subgroups of the metastatic phenotype are separated even further by changes in P53. It has been extensively researched how KRAS activation, the most common oncogenic mutation in pancreatic adenocarcinoma, contributes to the tumorigenic progression of pre-existing malignancies. The KRAS mutation is indicative of a poor prognosis in individuals with and without resectable malignancies, according to multiple studies (Ogura et al., 2013). According to functional studies, KRAS is essential for the advanced pancreatic cancer to continue growing. Modifications in the metabolism of cells have become apparent as targets for medical intervention. The higher metabolic needs of cancers generally necessitate coordinating the provision of nutrients. Because of the high hypoxia and restricted food availability caused by the hypo-vascular, fibrotic tumor microenvironment, pancreatic adenocarcinoma is particularly susceptible to metabolic stress. Thus, for pancreatic adenocarcinomas to grow and survive, a variety of acquired changes in nutrient uptake and utilization are necessary; many of these changes are regulated either directly or indirectly by oncogenic KRAS. KRAS stimulates the intake of external nutrients, such as glucose, and controls the scavenging of serum proteins and lipids through an endocytic mechanism called micropinocytosis (Commisso et al., 2013). Constitutively high levels of autophagy, which recycles organelles and protein aggregates by engulfing them in modified membranes called autophagosomes and breaking them down in lysosomes, are a feature of cancers containing KRAS mutations (Yang et al., 2011). By eliminating damaged components from the cell, autophagy can help detoxify it. It can also supply intermediate metabolites for energy production and biosynthesis that are produced from destroyed cargo. Research using human and mouse cells demonstrates that autophagy can be genetically blocked to prevent pancreatic cancer from growing, or it can be prevented by the drug chloroquine, which prevents lysosomal acidity and is presently being studied in clinical trials.

Surgical resection is still a crucial part of treatment with the goal of curing pancreas cancer, despite great progress being made in the investigation and development of several innovative systemic medicines. Therefore, new developments in surgical techniques could contribute to better long-term results for patients with pancreatic cancer. There has been a significant push to find modern improvements in the technical methods used for various aspects of pancreatic cancer surgery. The highest possibility of improving patient outcomes may be found in surgical resection, especially a R0 resection, and an appropriate surgical approach for patients with locally advanced pancreas cancer that is borderline resectable. Techniques that maximize vascular repair and surgical resection may produce the biggest gains in outcomes for this patient population (Schneider et al., 2021). Several studies have demonstrated that margin negative resection considerably improves survival (Tummala et al., 2013; Strobel et al., 2017). Advances

in surgical techniques have enabled more aggressive excision in conjunction with the repair of surrounding blood vessels. For patients with locally advanced illness, the well-established Appleby operation is a highly effective surgical approach. In addition to distal pancreato-splenectomy, this procedure involves the simultaneous removal of the common hepatic artery and celiac trunk. According to research, Appleby surgery may reduce the incidence of bleeding (Malinka et al., 2020). More research is needed to discover whether the Appleby approach improves clinical outcomes.

Mesopancreas surgery is another state-of-the-art surgical method for treating pancreatic cancer. Numerous studies have demonstrated the advantages of whole mesorectal excision in the management of rectal cancer. Surgeons are now considering treating pancreatic cancer with a similar anatomic architecture as a result of this success. The mesopancreas is a substantial, well-vascularized tissue that extends below the mesenteric arteries from the posterior aspect of the pancreatic head. Total mesopancreas resection has been suggested as a potential strategy to improve clinical outcomes because the mesopancreas is often the site of acceptable resection margins for patients with pancreas cancer involving the pancreatic head.

Whole mesopancreas resection was linked to a much higher percentage of margin negative resection, more resected lymph nodes, a lower recurrence rate, and a longer duration of disease-free life, according to a recent single-institution analysis (Xu et al., 2017). A bigger, more recent study carried out by a single institution revealed that 80% of all mesopancreas resections were due to tumor infiltration of mesopancreas fat, with a margin negative resection rate of almost 50% (Safi et al., 2021). Merely ten percent of patients who received margin negative resection experienced local recurrence, according to the multivariable analysis in this study, and only this type of resection was an independent prognostic predictor. The utilization of a "no-touch" methodology has improved this strategy even more.

Because physical manipulation of the contained tumor may cause tumor cell spilling into nearby tissues and blood vessels, the no-touch isolation technique has been studied for a range of disease sites, including surgery for cancer of the colon and the eye. A no-touch method for pancreaticoduodenectomy and total mesopancreas excision was created in order to lessen the possibility that pancreatic cancer cells would leak into the peritoneal cavity, portal vein, retroperitoneum, and other nearby anatomic structures. An evaluation of the clinical outcomes employing this no-touch approach in a retrospective analysis revealed an approximately seventy percent margin-negative resection rate. Patients who had margin-negative resection also had equivalent 5-year overall survival and recurrence-free survival rates of fourty two percent and thirty two percent, respectively, according to Hirota et al. (2017).

The irreversible electroporation technique uses high voltage, microsecond electrical pulses to cause cell membrane porosity, which eventually results in cell death. Patients with locally advanced pancreatic cancer have benefited from the surgical application of this technique. When compared to thermal ablation methods like cryoablation, microwave ablation, and radiofrequency ablation, infrared radiation treatment offers a number of benefits. These methods include minimal impact on blood flow, substantially sparing neighboring connective tissues, and

more precisely targeted tumor removal. Clinical studies indicates that the use of IRE significantly improves overall survival and both local and distant progression-free survival (Martin, 2015).

Moreover, patients undergoing evaluation for pancreatic cancer surgical resection are receiving more attention to clinical selection and medication management optimization. Thanks to improvements in surgical procedures, the post-operative mortality rate for patients receiving pancreaticoduodenectomy has declined, although the related morbidity rate is still very high.

A new predictor of post-operative morbidity following pancreatic cancer resection is sarcopenia. Sarcopenia was found to be a strong predictor of post-operative problems in a recent retrospective research. It was also linked to higher preoperative albumin levels and older age (Centonze et al., 2020). Moreover, research has looked at the body mass index (BMI) as a possible indicator of how well a pancreatic cancer resection will go. Longer post-operative hospital stays and shorter survival were linked to lower blood albumin levels and BMI, according to a sizable retrospective analysis (Hendifar et al., 2016). Immunotherapy may be able to lessen the immunosuppressive TME alterations. Immunotherapy has the ability to initiate an anti-tumor immune response by means of TME cell transformation. In recent years, a range of immune-modulating treatments have been used to treat cancer, such as cancer vaccines and checkpoint inhibitors based on monoclonal antibodies.

One of the most exciting advances in the treatment of modern cancer has been the use of checkpoint inhibitors, which have been around since the FDA approved ipilimumab, the first immune checkpoint inhibitor. At first, stage III/IV melanoma was approved for treatment with the medication ipilimumab. It works as an inhibitor of T-cell differentiation and IL-2 production by preventing the cytotoxic T lymphocyte antigen 4 (CTLA-4) from attaching to its ligands, B7.1 and B7.2ASDFGL (Serrels et al., 2015). Significant changes in T-cell immunosuppressive activity are also brought about by CTLA-4 activation, which triggers cellular pathways that generate regulatory cytokines and inhibitory proteins (Tao et al., 2021; Leach et al., 1996). According to Royal et al. (2010), CTLA-4 inhibition proved inefficient as a stand-alone treatment for stage III/IV PDAC, despite its efficacy in treating melanoma.

The field of healthcare is quickly embracing personalized medicine. At an exponential rate, methods for using each person's unique clinical, genomic, genetic, and environmental data to inform choices regarding illness prevention, diagnosis, and treatment are developing (Chan & Ginsburg, 2011). Moreover, the use of personalized biomarkers can help pharmaceutical companies increase the chance of success in their clinical trials. Personalized medicine makes it possible to provide therapies to subsets of patients who respond best based on their unique features (Lu et al., 2021). Biomarkers, whether for prognosis, prediction, or dosage selection, are becoming more significant in personalized medicine to determine the subset of patients who will benefit from a treatment. They might also help identify both positive and negative reactions to treatment, which would allow for an early assessment of safety and efficacy prediction. Therefore, biomarkers play a crucial role in both enabling personalized therapy and helping to choose which patients are most suited for treatment with particular medications (Landeck et al.,

2016). Biomarkers are regarded as molecular signals of increased benefit from or toxicity from a particular medication (Beckman and Chen, 2015); they are also known as a measuring variable linked to the course of a disease (Ballman, 2015).

Because there is a great deal of confusion around the distinction between prognostic and predictive biomarkers, this systematic review is particularly focused on identifying them using machine learning algorithms (Ballman, 2015). Many times, prognostic and predictive biomarkers are confused for one another. This can lead to ethical dilemmas for researchers and doctors as well as detrimental medical, economical, and moral consequences for patients. When a prognostic biomarker is mistakenly identified as predictive, therapeutic benefits for a subset of the population may be overestimated, leading to prescriptions for some people when they should be available to everyone. For instance, whereas immunotherapy was originally exclusively recommended for individuals with tumors that expressed PD-L1 significantly, more recent research indicates that immunotherapy may also be an effective course of treatment for patients whose tumor expression of PD-L1 is low (Hong et al., 2023).

The goal of machine learning is to build systems that can learn from experience, comprehend knowledge from accessible data, and think like humans do. This methodology works well for bioinformatics since the biological systems under study are extremely complicated. Furthermore, there are few theories in molecular biology research, and the majority of the time, experimental evidence is used. The discipline of molecular biology is ideal for machine learning techniques. One of the first areas where machine learning was used was in molecular biology. Stormo and associates used the perceptron approach to determine the initial translation sites in E. coli (Stormo et al., 1982).

In summary, the discipline of bioinformatics analysis has undergone a revolution with the incorporation of machine learning algorithms, which have improved biomarker discovery and clarified intricate biological processes. Among the many benefits of machine learning approaches are their capacity to work with high-dimensional data, identify non-linear relationships, and adjust to different types of datasets. Machine learning techniques make it possible to find subtle patterns and interactions in large-scale genomic, transcriptomic, and proteomic datasets, which aids in the identification of strong biomarker candidates with high prediction accuracy in the context of biomarker discovery. It is imperative to recognize the constraints of machine learning, such as the possibility of overfitting, the requirement for extensive and varied training datasets, and the difficulty of comprehending intricate models. Notwithstanding these difficulties, the research's methodology makes use of cutting-edge sequencing technologies and machine learning to provide a thorough way for figuring out the molecular makeup of pancreatic ductal adenocarcinoma (PDAC). This project intends to overcome current obstacles and open the door for the creation of tailored diagnostic and treatment plans for PDAC patients by utilizing the capabilities of both fields. This will ultimately improve clinical outcomes and advance precision oncology initiatives.

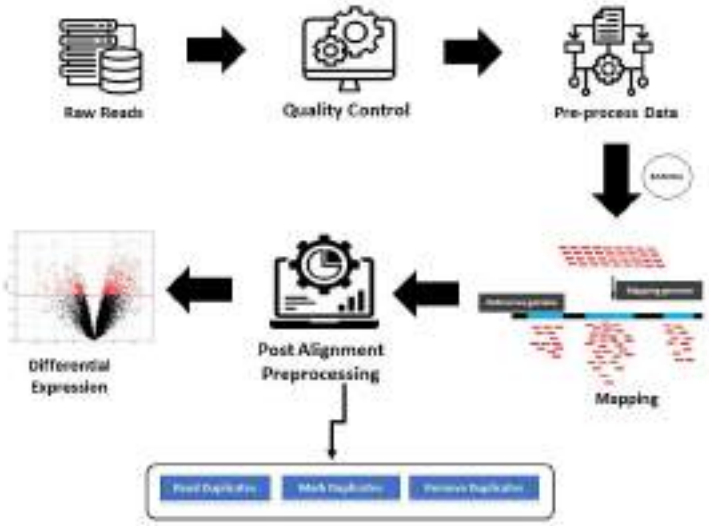
Chapter 3: Methodology

The identification of differentially expressed genes (DEGs) from many Microarray and RNA-seq datasets of PDAC is part of the study's overall workflow. Subsequently, common PDAC pathways were found, and potential biomarkers and treatment targets were discovered.

Inclusion criteria

Datasets were searched and retrieved from Array Express and EMBL-EBI by using key word pancreatic cancer and pancreatic ductal adenocarcinoma (PDAC). Selected datasets consist of *homo sapiens* origin, patients cell sample experiment excluding cell lines-based experimentation. Also, samples without any therapy, mutations or induced gene expression were selected for analysis. All datasets contain samples of PDAC and adjacent healthy pancreatic tissue region of patients.

Bulk RNA-Seq Analysis



3.1 Dataset Collection

For analysis dataset were selected on the basis of selection criteria, details of all dataset such as accession number, samples information and platform etc. Dataset with accession number E-GEOD-68086 comprises of 50 samples, out of which 25 were of PDAC and 25 adjacent normal samples.

3.2 Data Quality Evaluation & Preprocessing

Removal of technical noises in dataset prior to analysis is an essential preprocessing step.

Quantile normalization is widely used multi-sample normalization method. It was developed for RNA-Seq dataset normalization, now also used for the normalization of high-throughput data such as RNA-Seq and ChIP-seq (Hicks et al., 2017; Hicks and Irizarry, 2014).

PCA, Box Plot, and Relative Log Expression (RLE) Plot are used to evaluate the quality of normalized data. In high dimensional data, like microarray data, RLE plots are box plots that are used to display undesired variance from the gene median (Gandolfo and Speed, 2018). Next, duplicates were eliminated and low intensity probes were filtered to prevent noise in the data from non-specific hybridization of the probes.

3.3 Identification of differentially expressed genes (DEGs)

To find the genes with differential expression, a combination of the Bayes and Linear Regression models was used. To find differentially expressed (DE) genes in PDAC samples compared to normal, a linear model was fitted using the limma R package (Liu et al., 2021). (Ritchie et al., 2015). Then, the eBayes function in R was used to fit the "Empirical Bayes" model and compute the t-statistic, which yields the p-value for each differentially expressed gene (Klaus and Reisenauer, 2016). By using the p-value criterion of 0.05 and the logFC (log of fold change) thresholds (logFC < -1 and logFC > 1), significant DEGs were found for each dataset. The BH p-value or false discovery rate (FDR) of DEGs was likewise found to be less than 0.05. An upgraded volcano plot of DEGs was displayed or created, using logFC on the x-axis and -log10(p-value) on the y-axis, in order to show DEGs.

3.4 Preprocessing and Genome Alignment

To execute high quality analysis, quality assessment and preparation of sequencing data are essential steps. The RNA-seq readings were subjected to duplicate removal, base correction, and data quality assessment using the Fastp program (Chen et al., 2018). Fastp tool is developed in C/C++ language, and efficiently perform quality determination of sequence reads and then remove any detected sequencing errors like duplicates, low-quality bases, polyG and polyA tail, or adapter content (Chen et al., 2018). It combines the features of several tools, such as:

- Quicker than these stand-alone tools are FASTQC (Wingett and Andrews, 2018), Cutadapt (Martin, 2011), Trimmomatic (Sewe et al., 2022), and AfterQC (Chen et al., 2017). Sequence read mapping and alignment with the reference genome came after preprocessing. For the purpose of aligning reads with the human reference genome (hg38), Kim et al. (2019) developed the HISAT2 (hierarchical indexing for spliced alignment of transcripts 2) tool. With the graph Ferragina–Manzini (FM) index, it searches the genome quickly and is able to align reads from both DNA and RNA sequences. It is significantly faster than TopHat2, Bowtie, and BWA at performing splice alignment of whole genomes, transcripts, and exons (Liu et al., 2018). Next, an isoform reconstruction and transcript abundance calculation were performed using StringTie (Pertea et al., 2015), which produced a read count table. For effective transcript reconstruction, it makes use of the network flow method (Pertea et al., 2015).

- DEGs Identification Differential gene expression analysis was performed using Ballgown. Ballgown utilize read count table generated by String Tie (Pertea et al., 2015) and fit linear regression model to obtain differentially expressed (DE) genes (Pertea et al., 2015). Before fitting linear model on sequence reads, FPKM (fragments per kilobase of exon model per million mapped reads) normalization was performed to remove transcript length biases and sequencing depth in data (Zheng et al., 2011). Ballgown also help in visualization, and summarization of results. The statistically significant DEGs were filtered using the p-value, logFC, and FDR p-value thresholds, which were used for Micro Array datasets.

3.5 Raw Data Quality

We utilized a specific tool called FASTQC (Wingett and Andrews, 2018) to evaluate the data's quality, with scrutiny on the quality control report it generated. Each section of this report, focusing on the outcomes of individual FASTQC modules, offers unique insights. Apart from displaying graphical or listed data, it incorporates symbols denoting whether the data passed, failed, or raised a warning. We need to be cautious when interpreting these symbols to evaluate sequencing data, as they are assigned based on highly specific assumptions relevant to a limited set of circumstances.

3.6 Quality Control

To ensure the acquisition of high-quality data, it's crucial to rectify the deficiencies and subpar aspects pinpointed by FASTQC (Wingett and Andrews, 2018). This necessitates performing quality control and data preprocessing before employing RNA Seq tools. To fulfill this objective, we utilized Fastp, a program introduced in 2018 (version: 0.19.7). Fastp, written in C++, distinguishes itself as a swifter option compared to other preparation programs. Noteworthy is its comprehensive range of preprocessing functionalities, encompassing sequence trimming and formatting. The Fastp tool was employed to process the collection of raw FASTQC files, yielding two outcomes:

- An HTML quality assurance report
- A FastQ sanger file

Fastp was used with its default configuration, and the quality screening parameters were changed to meet the demands.

3.7 Second quality check of pre-processed data

FASTQC (Wingett and Andrews, 2018) was used once more for a quality assessment in order to determine whether the pre-processed readings' quality has improved. An assessment of the quality of these reads was done prior to putting them under more inspection.

3.8 Reference genome mapping

Transcripts are matched with a reference genome to perform an alignment method that pinpoints the raw readings with accuracy. HISAT2 (Galaxy Version 2.2.1+galaxy1), which is renowned for its accuracy and speed, is used in this procedure (Kim et al., 2019). Read alignment is performed using "Human GRCH38," the most recent human reference genome available in Galaxy, for both paired-end and single-end datasets. The orientation "Forward (FR)" was chosen for paired-end data. The number of transcripts that were successfully aligned to the reference genome is included in the post-alignment report that HISAT2 generates. The raw fasta measurements are aligned and then translated into the BAM (Binary Alignment Map) file format.

3.9 Read Duplication Identification and Removal

Both biological abnormalities and experimental artifacts can produce duplicate readings, which could affect the results. In order to reduce the possibility of false-positive results, it is essential to eliminate spurious duplicates produced by the Polymerase Chain Reaction (PCR) and incorporate biological replicates in order to reliably analyze gene expression differences.
This process of identifying and removing duplicates involves the utilization of two tools:

1. MarkDuplicates (Galaxy Version 2.18.2.3) (Zhao, 2018) and RmDup (Galaxy Version 2.0.1). Default parameter settings are applied to both tools. MarkDuplicates employs a unique flag that RmDup recognizes to flag duplicate readings.
2. Subsequently, RmDup identifies and eliminates these duplicate entries from the aligned BAM file.

3.10 Assembly and Quantification of Transcripts

StringTie (Pertea et al., 2015), version 1.3.4, a tool accessible on the Galaxy server, aids in discerning gene and transcript expression levels from aligned data files by consolidating them into potential transcripts. It relies on a reference annotation file for the human genome, known as hg38, for transcript assembly. The transcript files generated by StringTie are subsequently utilized for Ballgown analysis (Pertea et al., 2015).

3.11 Hub Gene Identification using Machine Learning

3.11.1 Data Loading

The RNA-seq data is loaded from a tab-delimited text file named 'expression_rnaseq.txt' into the R environment using the read.table() function. The header = TRUE argument indicates that the first row contains column names, as.is = TRUE preserves the character data type, sep = "\t" specifies the tab delimiter, and row.names = 1 sets the row names from the first column of the data.

3.11.2 Creating Level Factor

A factor variable Design is created, representing two levels ("low" and "high") for an experimental condition normal and disease. This factor is repeated to match the number of samples in the dataset.

3.11.3 Filtering Low Counts

The edgeR package (Robinson et al., 2010) is used to filter out genes with low counts. The mean log2 counts per million (CPM) for each gene across all samples are calculated. Genes with a mean log2 CPM greater than 1 are retained.

3.11.4 Normalization

To standardize the count data, Love et al. (2014) uses the DESeq2 software. Using the count data and the experimental design (~Design), a DESeqDataSet is first generated. Then, the regularized logarithm transformation (rlog()) is applied to obtain normalized counts. After normalization, another filtering step is performed based on the coefficient of variation (cofVar). Genes with a coefficient of variation greater than the 90th percentile are retained.

3.11.5 Defining Custom Score and Filter Functions

Custom score and filter functions are defined for feature selection using the caret package. These functions are specifically tailored for the selection process based on the performance of models.

3.11.6 Decision Tree Model (C 5.0)

The decision tree model is fitted using the C5.0() function from the C50 package (Ahmadi et al., 2018). The training data consists of the normalized counts, and the response variable is Design. Subsequently, a model summary is generated to examine the characteristics of the decision tree model, and a confusion matrix is printed to evaluate the model's performance on validation data.

3.11.7 Random Forest Model

Like the decision tree model, a random forest model (Rigatti, 2017) is fitted using the rfSBF control settings. The rfSBF control contains custom score and filter functions for feature selection. The performance of the random forest model is evaluated using a confusion matrix.

3.11.8 Results Analysis

Random seed (set.seed()) is set to ensure reproducibility of results. The performance metrics of both models (decision tree and random forest) are assessed using confusion matrices, which provide insights into the accuracy of class predictions.

Chapter 4: Results

4.1 Next Generation Sequencing Analysis

Following data retrieval from GEO, analysis is conducted using a suitable tools (given in section 3) within the Galaxy server to pinpoint DEGs. This process encompasses data pre-processing, transcript examination, and DEG identification.

4.2 Pre-processing of the Transcript Data

This dataset comprises a total of 50 samples, evenly split between PDAC patients and normal individuals. The raw data underwent analysis in Galaxy, where reads were globally trimmed at position 9 from the start. Following pre-processing, all samples were retained for further analysis, with transcript data analyzed using Ballgown in R. Figure 1 displays the distribution of transcript lengths, frequency on the y-axis and with transcript length in base pairs plotted on the x-axis. The distribution shows positive skewness with a right-tailed long tail, and the highest frequency occurs at a length of 100 base pairs.

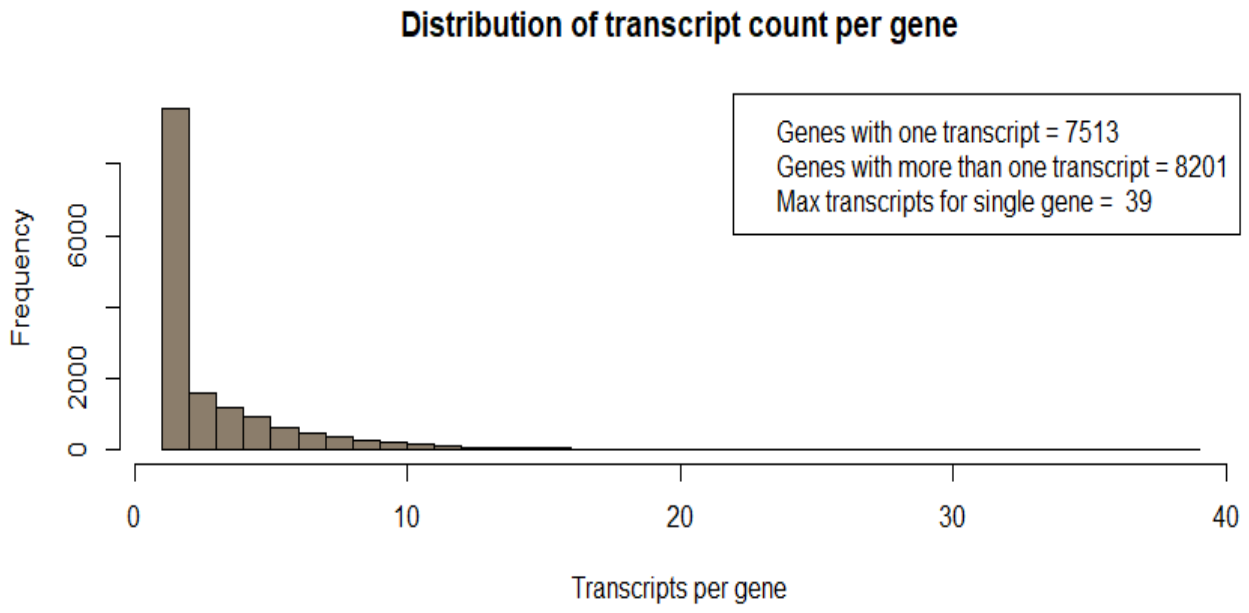


Fig1: *Distribution of transcripts per gene*

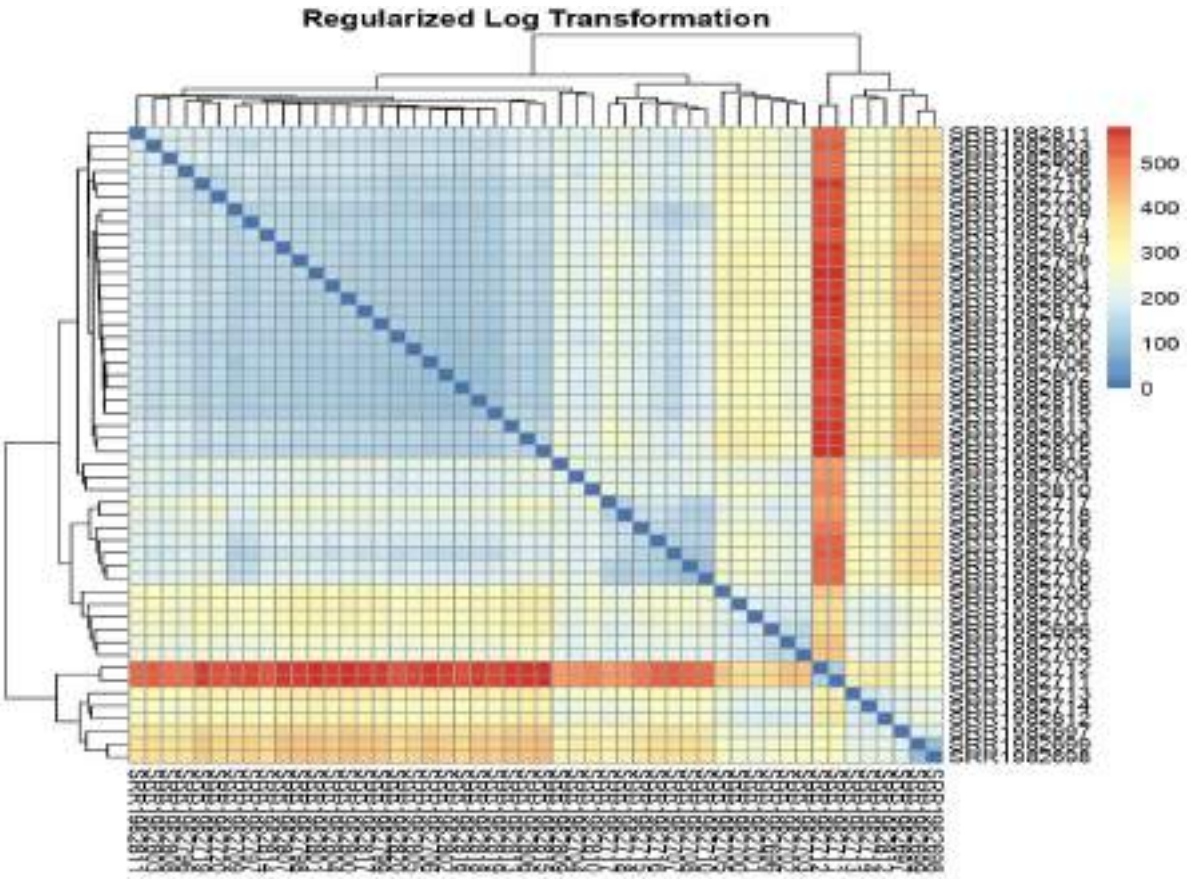


Fig2: *Regularize Log Transformation*

Figure 2 shows the changes in gene expression levels along a chromosome after the counts have been transformed using a regularized log function. Regularized log transformation is a mathematical process used to normalize RNA-seq data. RNA-seq data is used to measure the abundance of RNA molecules in a biological sample. Normalization is important because it allows researchers to compare gene expression levels between samples more easily. Overall, the graph shows the results of a regularized log transformation of RNA-seq data. The downward slope of the line suggests that the gene expression level decreases as the sequence progresses along the chromosome.

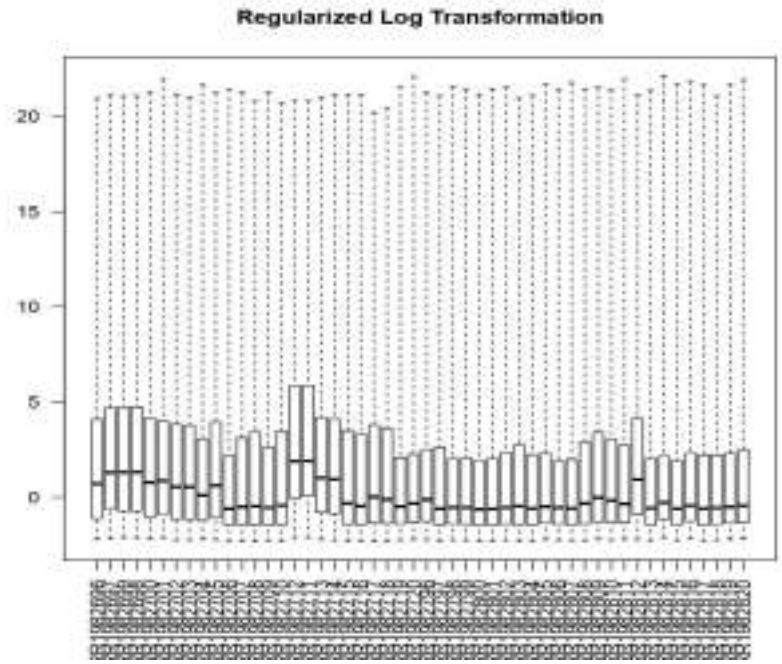


Fig3: *Box plot of regularize log transformation*

Figure 3 shows a few data points that are clustered around 0, and then a larger cluster of data points that are spread out between 0 and 5. There are also a few outliers that fall outside of these clusters. The fact that the data is plotted on a logarithmic scale suggests that the raw data values may vary over a wide range.

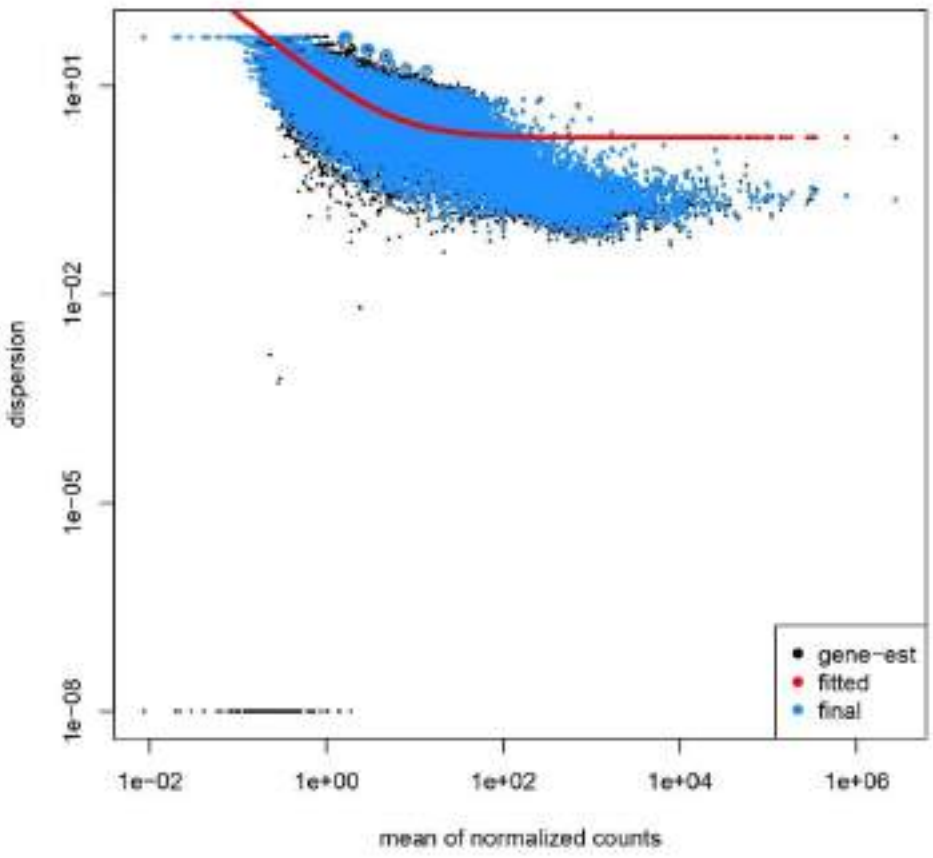


Fig4: *Dispersion plot of normalize counts*

Figure4 shows there are three lines plotted on the scatter plot. The black line labeled "fitted" a fitted curve showing the overall trend. The red points plotted with the label "final" may represent the actual data points. However, it is difficult to distinguish the red points from the fitted curve since they mostly overlap. The blue points with the label "dispersion" could represent outliers or data points that deviate from the fitted curve.

Overall, the plot suggests that there is a positive correlation between the number of genes and the frequency of mutations. Genes with a higher number of genes tend to have a higher frequency of mutations.

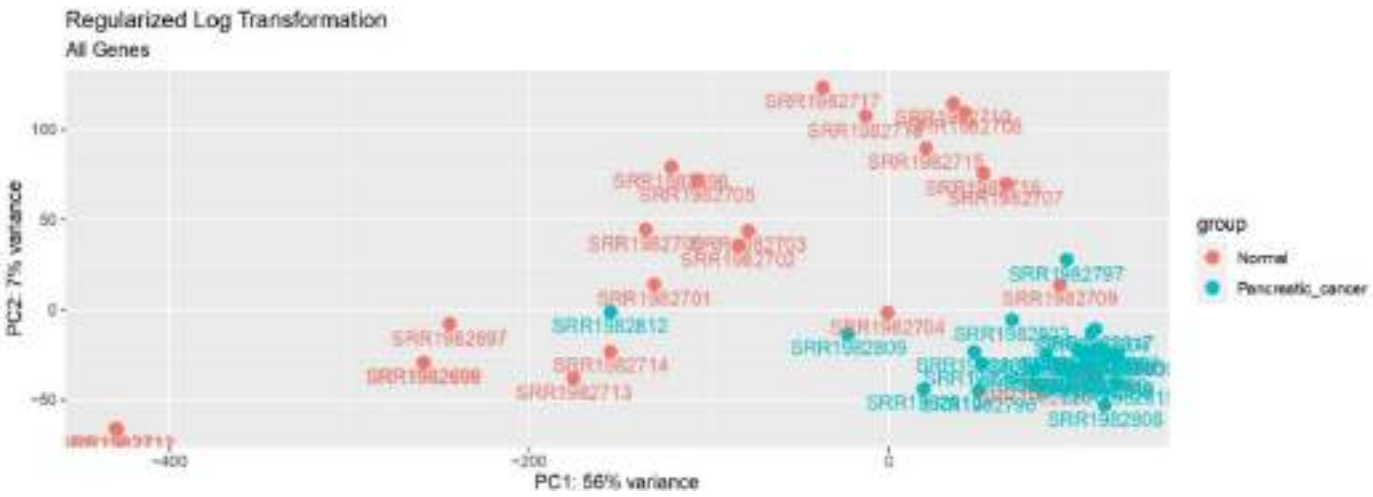


Fig5: *Scatter plot of Regularize Log Transformation*

In figure 5 scatter plot, the two variables are the percent of patients with a type of cancer in each group. The x-axis is labeled “PC1: 56% variance” and the y-axis is labeled “PC2: 7% variance.” Each data point on the scatter plot represents a different group. The label for each group is shown to the right of the plot. The group labeled "Normal" is clustered in the lower left corner of the plot, while the group labeled "Pancreatic_cancer" is clustered in the upper right corner of the plot. This scatter plot suggests that there is a correlation between the two variables.

4.3 Visualization of DEGs

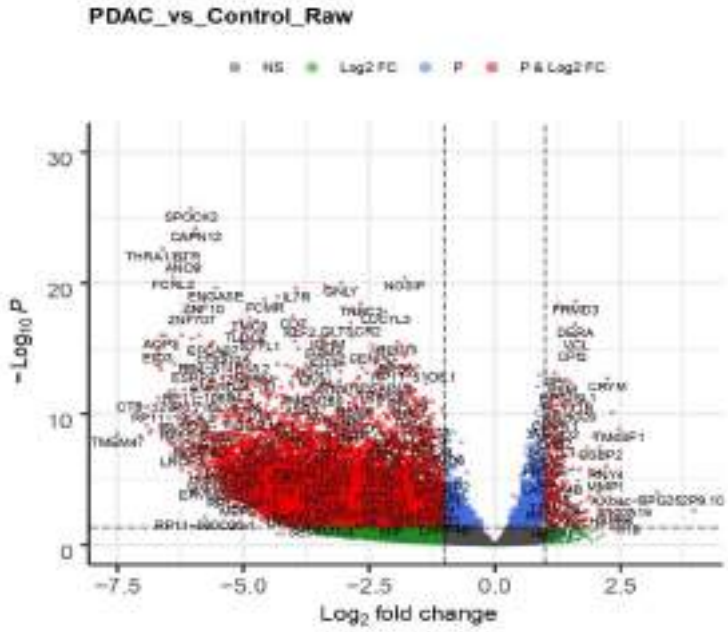


Fig6: *Volcano plot of Differentially Expressed Genes (DEGs)*

Figure 6 showcases an enhanced volcano plot, which visualizes the Differentially Expressed Genes (DEGs) based on their p-value and log2 fold change. The x-axis is scaled with the log2 fold change, while the y-axis displays the log p-value. Genes with a p-value <0.05 and a significant log2 fold change (>±1) are considered significant. Each dot on the plot represents a gene. Grey-dotted genes are deemed biologically and statistically insignificant. Genes marked with blue dots are statistically significant but biologically insignificant. Those with green dots are biologically significant but statistically insignificant. Red-dotted genes meet both criteria, being biologically and statistically significant, as they have a p-value <0.05 and a fold change greater than ±1.

4.4 Machine Learning

4.4.1 Data Preparation

In this phase, the RNA-seq data was preprocessed to ensure its suitability for downstream analysis. The data, stored in the file 'FullData.csv', was initially loaded into the R environment. To mitigate the impact of low-count genes, a filtering step was applied, resulting in a dataset comprising 10765 genes across 50 samples. Subsequently, normalization was performed using the DESeq2 package to account for differences in sequencing depth and library size. This normalization step yielded a refined dataset containing 3230 genes, which served as the foundation for subsequent analyses.

4.4.2 Feature Selection

Feature selection is critical for identifying the most informative genes that discriminate between the "high" and "low" categories. Two feature selection methods, namely random forest (rfSBF) and decision tree (dtSBF), were employed in this study. These methods aim to identify genes whose expression levels are most discriminative of the sample categories. FI=

Notably, attributes such as PRF1, ZNF200, and CD2 emerged as significant features based on their importance scores derived from the selection process. These features were subsequently utilized in model training to classify samples accurately.

```
C5.0 [Release 2.07 GPL Edition]         Thu May 16 13:43:53 2024
-------------------------------

Class specified by attribute `outcome'

Read 50 cases (101 attributes) from undefined.data

Decision tree:

PRF1 > 6.742264: low (23)
PRF1 <= 6.742264:
:...ZNF200 <= 2.67225: high (22)
    ZNF200 > 2.67225:
    :...CD2 <= 5.574034: high (3)
        CD2 > 5.574034: low (2)
```

Fig7: *Predicted significant features through machine learning model Decision Tree*

4.4.3 Model Training and Evaluation: Decision Tree Approach

In the decision tree approach, a C5.0 decision tree model was trained using the selected features to classify samples into "high" and "low" categories. The resulting decision tree exhibited a concise structure with a size of 4 nodes, indicating efficient utilization of discriminatory features. Notably, the model achieved perfect classification on the training data, with zero errors. However, evaluation on validation data revealed an accuracy of 76%. Further analysis of the confusion matrix highlighted a sensitivity of 64% and a specificity of 88%, indicating the model's ability to effectively distinguish between the two categories.

```
Confusion Matrix and Statistics

          Reference
Prediction high low
      high   16   3
      low     9  22

               Accuracy : 0.76
                 95% CI : (0.6183, 0.8694)
    No Information Rate : 0.5
    P-Value [Acc > NIR] : 0.0001529

                  Kappa : 0.52

 Mcnemar's Test P-Value : 0.1489147

            Sensitivity : 0.6400
            Specificity : 0.8800
         Pos Pred Value : 0.8421
         Neg Pred Value : 0.7097
             Prevalence : 0.5000
         Detection Rate : 0.3200
   Detection Prevalence : 0.3800
      Balanced Accuracy : 0.7600

       'Positive' Class : high
```

Fig8: *Confusion matrix and Statistics of Decision Tree Model*

4.4.4 Performance Comparison

A comparative analysis of the decision tree and random forest models revealed distinct performance metrics. While the decision tree model provided a straightforward interpretation with minimal complexity, it demonstrated relatively lower accuracy and sensitivity compared to the random forest model. In contrast, the random forest model exhibited superior accuracy and sensitivity, suggesting its efficacy in handling complex classification tasks involving high-dimensional data such as RNA-seq. Overall, the results underscore the importance of selecting appropriate machine learning algorithms to maximize classification accuracy and predictive performance.

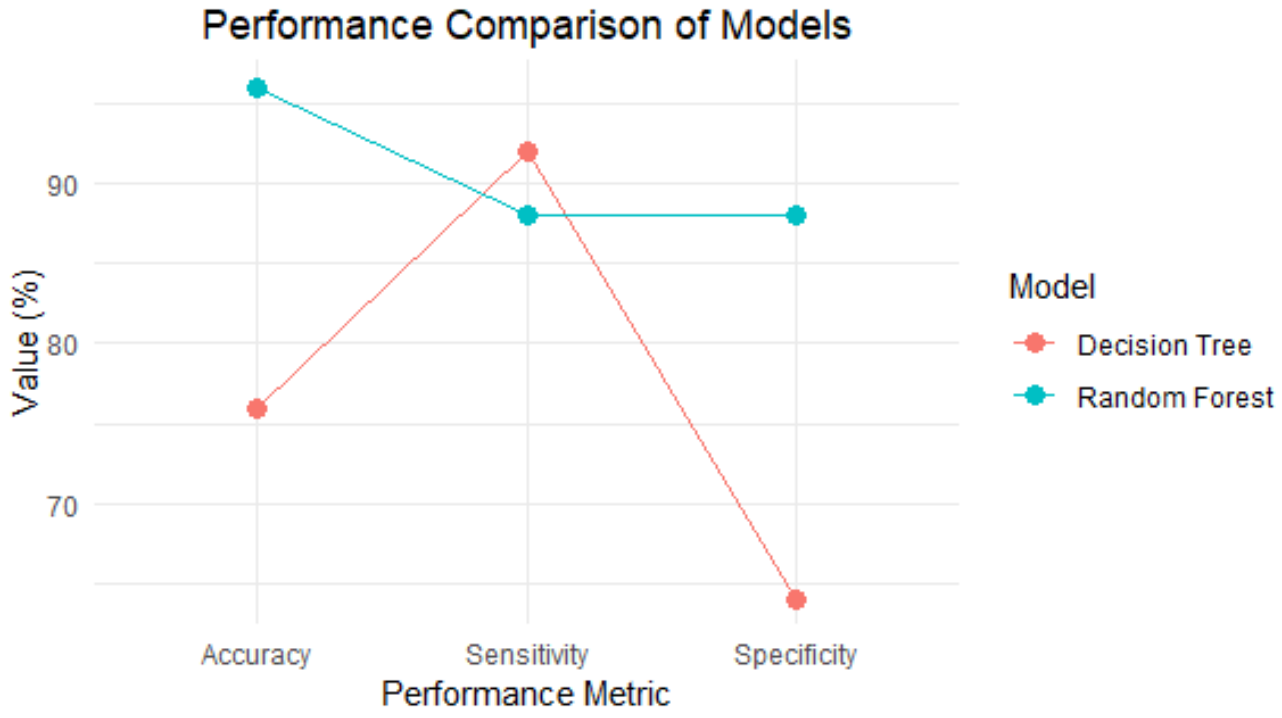


Fig9: *Machine Learning Models Performance Comparison on PDAC Samples*

In **Fig9**, Random Forest model is consistently higher than the decision tree model on performance as based on the graph its suggesting that it has fewer mistakes, it's missing fewer PDAC cases and the model is avoiding misclassifying normal samples as PDAC.

This graph depicts the performance of decision tree (red line) and random forest (blue line) models in classifying PDAC samples. The y-axis represents various performance metrics (unspecified due to missing labels). Higher values generally indicate better performance. Based on the graph, we can observe that one model performs better than the other across all metrics (though specific conclusions require knowing the metric labels). This comparison aids in selecting the optimal model for PDAC sample classification in this study.

Fig10: *Performance Comparison of Machine Learning Models for PDAC Classification*

Fig 9 is a bar graph comparing the performance of two machine learning models, a decision tree and a random forest, on a classification task. The y-axis shows the accuracy, represented as a percentage, which indicates how well each model distinguished between PDAC and normal samples. The x-axis represents the two models themselves (Decision Tree and Random Forest). Based on the graph, the Random Forest model (blue bar) appears to have a higher accuracy (around 75%) compared to the Decision Tree model (red bar) which has a lower accuracy (around 50%).

4.5 Identified Hub Genes

Notably, attributes such as PRF1, ZNF200, and CD2 emerged as significant features based on their importance scores derived from the selection process. These genes, often referred to as "hub genes" due to their central role in discriminating between sample categories, exhibit pronounced expression differences across different experimental conditions. These hub genes were deliberately added to the model training procedure to effectively categorize data into their appropriate groups by utilizing their discriminatory ability. The analysis focused on these salient characteristics in an attempt to clarify their possible biological importance and their role in the observed phenotypic variances. The significance of these hub genes in successfully differentiating between "high" and "low" categories was highlighted by further model evaluation and validation, underscoring their applicability as prospective biomarkers or targets for additional research in the context of the biological system under study.

Chapter 5: Discussion

The fourth most common cause of cancer-related deaths is pancreatic cancer. According to Kirby et al. (2016), this disease develops when a pancreatic cell sustains injury that sets off the unchecked growth of cancerous cells. Based on GLOBOCAN 2018 forecasts, there is anticipated to be a significant increase in pancreatic cancer mortality (79.9%; 3,45,181 fatalities) and incidence (77.7%; 3,56,358 new cases) between 2018 and 2040. This pattern is explained by the limited influence of preclinical and clinical research over the past fifty years, which has not improved patient outcomes or improved treatment and prevention strategies. The key to reducing this is developing new treatment strategies and putting primary and secondary preventative measures in place to lower the incidence and death rates of pancreatic cancer (Casolino et al., 2021). Exocrine and neuroendocrine are the two primary subtypes of pancreatic cancer. According to Lu et al. (2017), exocrine pancreatic ductal adenocarcinoma is the most common kind, accounting for over 85% of cases. People 65 years of age and older have a higher chance of getting pancreatic cancer (Kirby et al., 2016). Pancreatic cancer is caused by a variety of hereditary and acquired genetic changes. Its development is significantly influenced by inherited genetic variants with high and low penetrance. Patients with pancreatic genetic alterations may benefit more from tailored drugs because they may react better to individualized care (Chen et al., 2017). According to Lu et al. (2017), patients with distant metastases have a worse prognosis, with a median survival time of 3 to 6 months, compared to 8 to 12 months for those with locally advanced malignancy.

Early diagnosis of malignancies is crucial for enhancing treatment regimens that aim to manage the disease efficiently, lengthen disease-free survival, and improve patients' overall quality of life. Systemic chemotherapy is usually followed by decisive and frequently curative therapies such surgery, ablation methods, or three-dimensional intensity-modulated radiation if tumors have been identified. All genes and their isoforms may be comprehensively annotated and quantified across datasets thanks to high-throughput RNA sequencing, or RNA-seq, providing a full picture of the transcriptome. RNA-seq has several advantages over microarrays, including reduced technical variability, increased resolution, a wider detection range, and the capacity to study new transcripts (Corchete et al., 2020). As such, it is not only wise but also necessary to actively test for pancreatic ductal adenocarcinoma (PDAC) at an early stage. CA19-9 remains the most widely utilized blood-based biomarker for PC diagnosis, prognosis, and treatment that has received FDA approval. However, it has some shortcomings, including low specificity, lack of expression in the Lewis negative phenotype, and a higher chance of false positive elevation in the presence of obstructive jaundice (Ballehaninna and Chamberlain, 2012). For pancreatic cancer management, numerous putative biomarkers have been found, such as cytokeratin, glycoprotein, mucinic indicators, hepatocarcinoma-intestine-pancreas protein, and PC-associated protein markers (Ballehaninna and Chamberlain, 2013), but none have been able to outperform CA19-9 in validation cohorts. A novel tissue biomarker based on five genes for PDAC diagnosis was previously developed by researchers using an inventive meta-analysis strategy across different transcriptome studies (Ballehaninna and Chamberlain, 2013).

This biomarker panel, at the tissue level, demonstrated the ability to distinguish pancreatic ductal adenocarcinoma (PDAC) from non-tumor tissue, PDAC precursors, chronic pancreatitis (CP), other malignancies, and healthy controls with a sensitivity of 94% and specificity of 89% (Bhasin et al., 2016). However, challenges associated with obtaining biopsy samples raise questions about the relevance of tissue-based diagnostic indicators. Moreover, much of the recent research relies on small sample sizes, limiting their capacity to identify dependable biomarkers. The primary unmet need for early PDAC

diagnosis remains the absence of reliable blood-based biomarkers, given the unpredictable nature of pancreatic cancer.

Several genome-level studies have been driven by the pressing demand for enhanced PDAC diagnostics, aiming to uncover new treatment targets and biomarkers for early detection. Despite extensive genomics research, a dependable biomarker surpassing CA19-9 in terms of sensitivity and specificity remains elusive. The disparity in datasets stemming from various sources, platforms, methodologies, and analytical techniques, coupled with a lack of agreement, inherently constrains independent research due to statistical limitations (Ramasamy et al., 2008). When conventional procedures are used, the gene signatures published in individual microarray studies frequently do not match those of comparative and meta-analysis studies, mainly because of discrepancies in analytical methodologies (Ramasamy et al., 2008).

RNA-seq analysis, machine learning, and Next Generation Sequencing (NGS) were utilized in this multidisciplinary study to explore the molecular landscape of PDAC and find possible biomarkers and therapeutic targets. Our research provides insight into important genetic changes and signaling networks connected to the pathophysiology of PDAC, opening up new possibilities for the development of precision medicine and targeted treatment strategies. Our research demonstrates the usefulness of RNA-seq analysis in deciphering the complex transcriptome profile of PDAC. Genes with notable alterations in expression were identified using differential expression analysis between PDAC samples and normal controls. A thorough overview of differentially expressed genes (DEGs) was made possible by visualization techniques including volcano plots, which made it easier to identify biologically significant candidates for additional research. The identification of DEGs is crucial for understanding the molecular basis of PDAC and identifying potential targets for therapeutic intervention.

The integration of machine learning algorithms in bioinformatics analysis has emerged as a powerful tool in cancer research, enabling the identification of biomarkers with high predictive accuracy and elucidating complex biological processes. In line with previous literature, our study underscores the importance of machine learning in biomarker discovery. By leveraging machine learning techniques, we identified significant features such as PRF1, ZNF200, and CD2, which played pivotal roles in distinguishing between PDAC and normal samples. These hub genes, deemed "biologically and statistically significant," offer valuable insights into the molecular mechanisms underlying PDAC pathogenesis and may serve as potential biomarkers for diagnostic and prognostic purposes.

However, it is essential to acknowledge the limitations of our study. While machine learning techniques offer several advantages in biomarker discovery, including their ability to handle high-dimensional data and uncover subtle patterns, they are not without challenges. Overfitting, the need for large and diverse training datasets, and the interpretability of complex models remain significant concerns. Additionally, the retrospective nature of our study limits causal inference and generalizability.

Chapter 6: Summary

Pancreatic cancer is the fourth leading cause of cancer-related deaths, developing from unchecked malignant cell growth in the pancreas. According to GLOBOCAN 2018, pancreatic cancer mortality and incidence are expected to rise significantly between 2018 and 2040 due to the limited impact of past research on patient outcomes and treatment strategies. New treatments and preventative measures are crucial to reducing these rates. Pancreatic cancer has two main subtypes: exocrine and neuroendocrine, with exocrine pancreatic ductal adenocarcinoma (PDAC) being the most common, especially in individuals over 65. Various genetic changes contribute to its development, with inherited genetic variants influencing its onset. Tailored treatments for patients with specific genetic mutations may improve outcomes. Early detection is vital for effective treatment, extended disease-free survival, and better quality of life. High-throughput RNA sequencing (RNA-seq) is superior to microarrays for comprehensive gene analysis, offering detailed transcriptome insights. Despite the widespread use of the CA19-9 biomarker for PDAC, it has several limitations, prompting the search for better biomarkers. A tissue-based biomarker panel demonstrated high sensitivity and specificity for distinguishing PDAC from other conditions, but biopsy challenges and small sample sizes limit its application. Reliable blood-based biomarkers are essential for early PDAC diagnosis.

Genome-level studies have sought to improve PDAC diagnostics and treatment targets, but no biomarker has yet outperformed CA19-9. Discrepancies in datasets and methodologies further complicate biomarker validation. This study utilized RNA-seq, machine learning, and Next Generation Sequencing (NGS) to explore PDAC's molecular landscape, identifying potential biomarkers and therapeutic targets. Differential expression analysis highlighted key genes and pathways involved in PDAC, offering new avenues for precision medicine. Machine learning algorithms enhanced biomarker discovery, identifying significant genes such as PRF1, ZNF200, and CD2, crucial for distinguishing PDAC from normal samples. These findings provide insights into PDAC pathogenesis and potential diagnostic markers. However, the study's limitations include overfitting risks, the need for extensive training datasets, and the interpretability of complex models. The retrospective nature of the study also limits causal inference and generalizability.

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